1 Methods

We summarize each step of Figure 1 in the figure's row order, and the Definitions section defines the terms. Ticket counts are unplanned tickets unless stated otherwise. We rebuilt the midterm analysis's case table from IPA's service case export, kept its rule for unplanned tickets, and linked the tickets by exact serial number to the clearView telemetry database and to IPA's installed-base list, which supplies the fleets behind every rate per installed machine-month. IPA's answers on Oct 13th set how we read ticket times and fields. Its two questions on slides 4 and 5 set the follow-ups, and its requests on Oct 30th set the Electronics case list and a redrawn chart for slide 4a.

Customer fleet ownership. We counted the telemetry devices that ticket serials reach and compared accounts with sites.

High-ticket machines. We flagged rate outliers and chronic outliers [one-sided exact Poisson test; Benjamini–Hochberg], compared ticket concentration with chance [Poisson simulation], re-tested the rate outliers allowing for spread between machines [negative binomial], and compared the rise in non-success codes before tickets on chronic outliers and other machines.

Repeat tickets. We measured the share of eligible tickets followed by another ticket on the same machine within 30 days and within 7 days, and the 30-day share without monitoring tickets. With the follow-ups, we compared the first two with random timing [exact calculation; Monte Carlo check].

Rates by model and age. We computed rates per 100 installed machine-months by model, age, and service status and compared each model with the fleet raw and age for age [indirect standardization; Poisson main-effects model; iterative proportional fitting; cluster bootstrap].

Customers vs their fleets. We compared each customer's tickets with its fleet's expected tickets [one-sided exact Poisson test; Benjamini–Hochberg; overdispersion-adjusted funnel plot] and compared ticket concentration with chance [Poisson simulation].

Persistence and outliers. We split the period into two halves, refitted expected tickets in each, and compared the halves [Spearman rank correlation; stratified permutation test]. We flagged consistent outliers [one-sided Poisson tests; Benjamini–Hochberg] and machines above their own expected tickets beyond chance [one-sided exact Poisson test; Benjamini–Hochberg; negative binomial].

Ticket clock and fields. We checked whether opening times step backwards in case-number order on the export's clock and on hospital-local clocks. We compared how often the issue fields are filled on open tickets and on August 2026 tickets with status Trouble Closed.

First look at codes. For the paired tickets, we compared the last 10 codes before each ticket with those in its comparison window by code group, with no test.

Association results. Following a plan and a Category-to-code crosswalk fixed in advance, we compared the odds that a ticket's own codes appear in its final 7 days rather than in the final 7 days of its comparison window with the same odds for other faults' tickets. The ratio is adjusted for machine type and quarter and pooled over the code sets that could be tested [Mantel–Haenszel odds ratio; one-sided permutation test; cluster bootstrap]. We then ran planned sensitivity checks.

Request types by age. To answer whether young machines' extra tickets are different kinds of problem, we compared rates on young and older machines by request group, all models together [cluster bootstrap], and compared the mix of request groups within model [permutation test; Benjamini–Hochberg; Mantel–Haenszel share ratio]. We split the extra tickets into within-model and model-mix parts and, after seeing the data, grouped machines by install year.

Repeat-contact pairings. To answer what follows what in repeat contacts, we classified each repeat and compared same-Category shares by the gap between a machine's tickets. We compared each model and telemetry type that has at least 100 eligible tickets with random timing, relative to the fleet [cluster bootstrap; Benjamini–Hochberg].

Electronics cases. As a descriptive step outside the plan, we compared Electronics and its Categories on young and older machines, all models together, and examined Touchscreen tickets by Sub Category, model, and install date. We built a case list, which an independent rebuild checked.

Verification. We recomputed every case-export number of the midterm analysis three independent ways and re-ran its telemetry results on the full database. The association study and the answers to IPA's two questions followed plans fixed in advance, with later changes recorded as dated amendments. We checked each analysis with a review or an independent rebuild, written from the definitions or plan without reading the analysis code. The exception is the Oct 14th first look at codes, which the association study later reproduced.

2 Results

Customer fleet ownership: Ticket serials reach 3,041 of 7,183 devices (42%), and accounts are individual facilities (845 accounts at 725–837 sites), so fleets come from the installed-base list [1].

High-ticket machines: We found 113 chronic outliers at 82 customers. The top 5% of the 6,621 telemetry devices hold 32.9% of their tickets vs. 22.2% by chance. Allowing for spread between machines, only 2 rate outliers still stand out, so high-ticket machines form a continuum. Non-success codes rise before their tickets no more than on other machines [2].

Repeat tickets: Of 6,665 eligible tickets, 23.6% repeat within 30 days and 9.5% within 7 days, 1.53 and 2.30 times random timing's shares. Without monitoring tickets, the 30-day share is 22.0% [3].

Rates by model and age: The fleet runs at 6.2 unplanned tickets per 100 installed machine-months. The three busiest models (LVD-Remote-2.0, LVRR-WP-2.0, and MV-2.0) hold 42% of machine-months and 51% of tickets, partly because they are younger. scrubBank 480 runs 2.6 times the fleet age for age. Machines under 4 years run about 8 per 100 vs. 2.5 at 10 years or more, but the 10+ rate overstates the drop with age: 83 of its 164 machines with a ticket are Not Serviceable, and removed machines are missing from the list [4].

Customers vs. their fleets: 63 customers exceed their fleets beyond chance (1,732 tickets vs. 747 expected), and 1 lies outside the normal spread. The top 10% of customers hold 47% of tickets vs. 39% from their fleets plus chance [4].

Persistence and outliers: Customers above their fleets in the first half tend to stay above them in the second (rank correlation 0.46 vs. 0.04 when shuffled), and machines with 3 or more first-half tickets had 2.6 times their expected tickets in the second half. 18 customers and 36 machines are consistent outliers. Of the 106 chronic outliers on IPA's list, 93 stay flagged, and no flagged machine stands out as a separate group [4].

Ticket clock and fields: Opening times never step backwards in case-number order (0 of 7,470 steps) but would 1,295–1,312 times on hospital-local clocks, so case times use one U.S. Eastern clock, as IPA confirmed. Trouble Code and Category are filled on 1 of 64 open tickets and on nearly all of the 461 August 2026 tickets of any kind with status Trouble Closed. Category is set near close, and the export holds only each field's final value [5].

First look at codes: Door codes appear before 68% of the 3,259 paired tickets vs. 61% of their comparison windows, and electronics codes before 20% vs. 13% [6].

Association results: The odds that a ticket's own codes appear in its final 7 days rather than in its comparison window's final 7 days are 3.2 times those of other faults' tickets (95% interval 2.3–4.8). By code set, the ratios are rough and do not rank the four sets with ratios above 1: Main Door is borderline, and Computer Board was not shown to be specific. At 7 days, the pooled result holds in every sensitivity check run so far that gives one, but it rests almost entirely on scrubex machines and is provisional [6].

Request types by age: Young machines run 8.0 tickets per 100 machine-months vs. 4.8 for older ones (1.66 times), with higher rates in nearly every request group. Within model, Electronics' share is 1.31 times as large on young machines (mostly Touchscreen) and Mechanical's 0.70 times. 80% of the 1,447 extra tickets come from young machines running above older ones of the same model. Touchscreen tickets cluster on 2022–23 installs (1.54 per 100, 3.9 times the 2024–26 rate) [7].

Repeat-contact pairings: Of the 1,574 repeats, 31% are the same Category, 10% the same request group with another Category, and 32% a different group. The other 28% involve monitoring, No Trouble Found, or a blank Category. Over all pairs of a machine's tickets, 46% of pairs a day or less apart share a Category vs. 27% of pairs over six months apart. No model or telemetry type with 100 or more eligible tickets clearly stands out from the fleet [3].

Electronics cases: Per machine-month, Electronics runs 1.71 times as often on young machines, near the 1.66 for all tickets. Touchscreen stands out at 4.2 times: of its 411 tickets on young machines, 344 are connector reseats or replacements, 402 are on the four 2.0 models (LVD-Remote-2.0, MV-2.0, LVRR-WP-2.0, and SV-48-2.0), and 315 are on 2022–23 installs. Its rate steps from 0.36 to 1.61 per 100 between Jan–Apr and May–Aug 2022 installs, which are nearly the same age, so the step points to a production, part, or installation change. The case list has all 1,605 unplanned Electronics cases on the fleet machines behind slide 4 [7].

Verification: Every case-export number reproduced exactly, and on the full database 65 of 66 checks passed; the one failure was a sort-order bug in our comparison script. Independent rebuilds matched the 113 chronic outliers, the fleet rates before a statistical review's six changes, all 33,515 compared values of the association study, the follow-ups' headline numbers, and every sheet of the case list. Reviews found that the chronic-outlier count runs from 62 to 133 with the statistical cut-off. The association results' 45 review points were all upheld and applied, and no headline result changed [8].

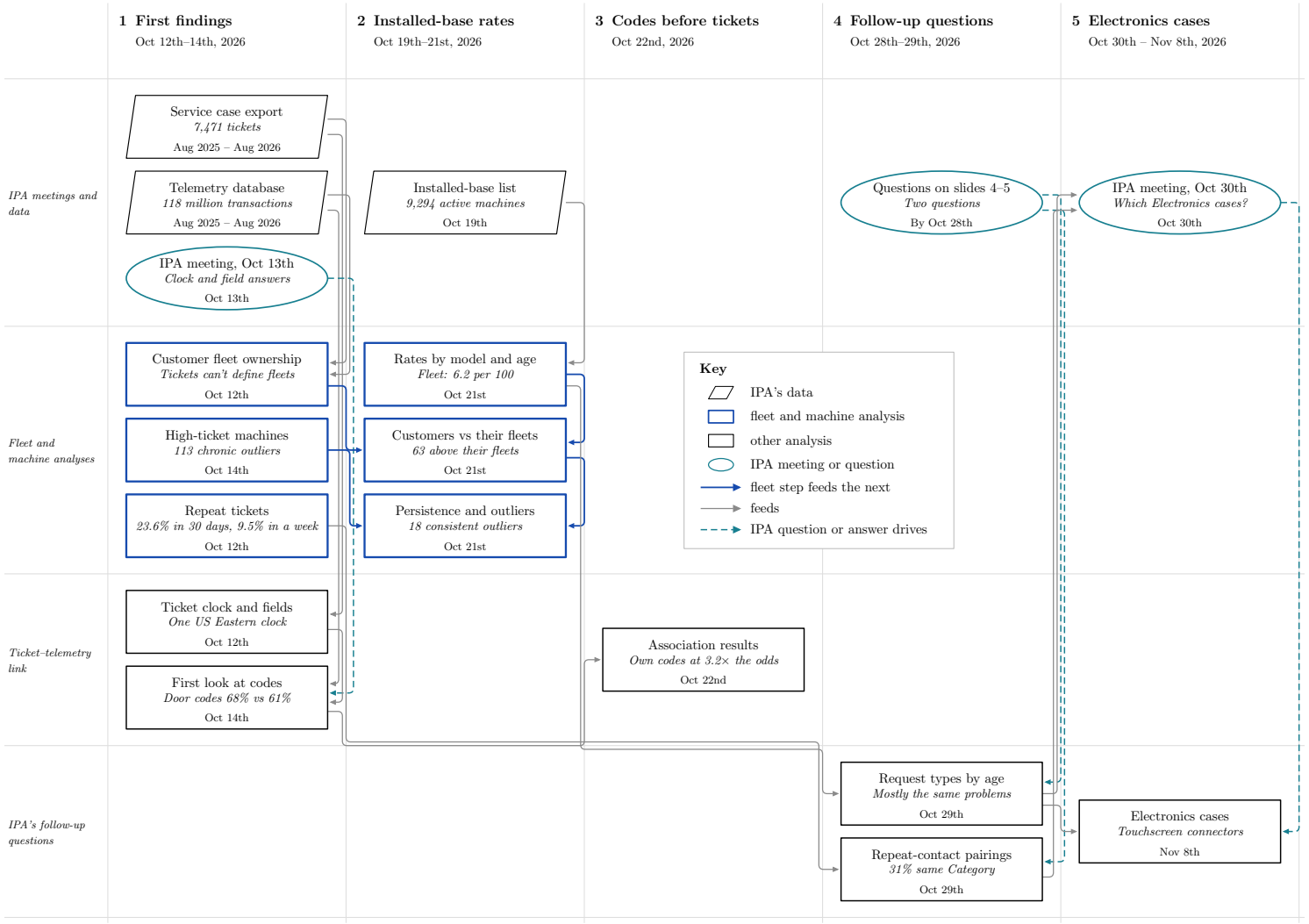


Figure 1: How the IPA analysis fits together, October 12th to November 8th, 2026. Rows are kinds of work and columns are phases in time order; each arrow shows what a step took from an earlier one. Cobalt marks the fleet and machine analyses and the arrows between them, grey arrows feed a step, and dashed teal arrows carry an IPA question or answer. To keep the chart readable, each of IPA's data sources has arrows only to the steps that use it most directly.

Definitions

We define the terms used in the Methods and Results, grouped by the rows of Figure 1. Statistical methods are named in brackets.

IPA meetings and data

Service case (ticket): One row of IPA's case export, one Case Number: 7,471 cases opened Aug 1st, 2025, to Aug 31st, 2026.

Unplanned ticket: Every case except three kinds, leaving 7,186:

- Planned work: Trouble Code Preventative Maintenance or Installation, or Case Record Type Install Case (23)
- Canceled requests: Category Cancelled Request (247)
- Duplicates: Category Duplicate Case (15)

No Trouble Found, blank-Trouble-Code, and monitoring tickets are considered.

Trouble Code, Category, and **Sub Category:** The export's three issue fields. It only holds each field's final value.

- **Trouble Code** is the top level (10 values, such as Electro/Mechanical, Electronics, and Network). IPA said it is entered at intake and may change by close.
- **Category** (41 values, such as Deposit Door, SAM, and Touchscreen) is set near close.
- **Sub Category** is also set near close and mostly records the repair action.

Account, Machine, and **Model:**

- **Account** is the case's Account Name (845, each with one City, State, and ZIP address).
- **Machine** is a serial number. It is the Serial # on the Salesforce case history, the Serial Number on the fleet list, and the device ID in telemetry. We matched them as exact text after trimming spaces. 3,773 serials have tickets.
- **Model** is the case's Model # (31 models) or the list's Model (34). They agree on all 7,018 matched tickets.

The install-base list, matched: IPA's active equipment list as of Oct 19th, 2026: 9,294 machines at 1,066 accounts, with 34 models.

- It holds only machines active on that date, so machines removed during the period are missing (433 unplanned tickets on 226 serials).
- A ticket is matched when its serial is on the list: 7,018 of 7,471 (93.9%).

The telemetry database, device, codes, and **telemetry type:** clearView's transaction log: 118,047,569 transactions from Aug 1st, 2025, to Aug 28th, 2026, each with a time and one of 49 codes.

- **Device** is a row of the device table (7,183); 6,951 logged a transaction.
- **Codes:** 00 is OK. Any other code is **non-success** (6.7% of transactions, mostly stock-outs and door events), so non-success doesn't mean a fault.
- **Telemetry type** is the device's clearView machine type (blank = Unknown).

Midterm analysis: The SQL notebook of Oct 10th, 2026, on the case export and the full clearView database. Our work began by reproducing it.

Fleet and machine analyses

The fleet and **a customer's fleet:** The fleet is every machine on IPA's list with installed time in the period, except the 239 on the third-party Vogeltronix plan, whose service rarely reaches this export, and the 109 installed after the period: 8,946 machines and 109,621 installed machine-months. A customer's fleet is its machines in the fleet. In the repeat comparisons, the fleet is all 6,665 eligible tickets.

Accounts are individual facilities: The 845 accounts sit at 725 distinct City, State, and 5-digit ZIP combinations, or 837 distinct addresses. Parent health systems are not in the data.

Top 5% by unplanned tickets: Rank the 3,685 machines with an unplanned ticket in the export. The cut is the 185th machine's count, 5 tickets, ties included: 247 machines.

Rate outlier: A device with improbably many unplanned tickets (Aug 1st, 2025–Jul 31st, 2026) for its type's rate × its months active [one-sided exact Poisson test; Benjamini–Hochberg $q < 0.05$]. Months active are the calendar months from its first to its last transaction in that window. Devices on IPA's shared internal host or with no host are removed, leaving 6,621.

Chronic outlier: In the top 5% and a rate outlier: 113 machines at 82 customers. We use it as a screening flag.

Rise in non-success codes: Take non-success codes per 1,000 transactions in the 30 days before each unplanned ticket that has a clean comparison window. The rise is the median of those windows minus the median of the comparison windows: +20.3 for chronic outliers, +21.6 for other top-5% machines, and +21.5 for all other machines.

Installed machine-month: For a fleet machine: time from the later of Aug 1st, 2025, and its install date to Sep 1st, 2026, in months of 30.4375 days. Partial months count.

Raw and **age for age:** A model's raw ratio is its rate ÷ the fleet's. Age for age divides its tickets by those its machines would have at the fleet's age-band rates [indirect standardization; Poisson main-effects model].

Machine age and **under 4 years:** Age is the years from install to the midpoint of its installed time in the period, or of each half for the two halves. Under 4 years (young) means installed after about Feb 15th, 2022.

Customer and **expected tickets:** A customer is the machine's account on IPA's list (for chronic outliers, the account named most often on its tickets). A machine's expected tickets are its machine-months × the fleet's base rate × a model factor × an age-band factor [Poisson main-effects model], and a customer's are the sum over its machines.

Beyond chance and **normal spread:** Beyond chance means at least 5 tickets and a count improbably high for the expected tickets [one-sided exact Poisson test; Benjamini–Hochberg $q < 0.05$], and we use it as a screening flag. Outside the normal spread means above the 99.8% limit of a funnel plot widened by the spread between customers [overdispersion-adjusted funnel plot].

The two halves: Aug 1st, 2025, to Feb 14th, 2026, and Feb 15th to Aug 31st, 2026. The rank correlation covers the 311 customers expected to have at least 3 tickets in each.

Consistent outlier: Flagged beyond chance under at least 10 of 12 variations of the method and above expected in each half [one-sided Poisson test, $p < 0.05$]. The variations cross three choices of expected tickets, tickets or 7-day episodes (runs with no gap over 7 days), and all tickets or tickets without monitoring and No Trouble Found.

Eligible ticket and **repeat:** An eligible ticket is an unplanned ticket opened before Aug 2nd, 2026, so it has 30 days of the export after it. It is a repeat when the machine's next unplanned ticket opens within 30 days. A repeat contact is not necessarily a failed repair.

Monitoring ticket: A Monitoring – Daily/Poll template ticket, found from its Trouble Reported text (138 tickets of any kind). The 22.0% removes them from each machine's sequence.

Random timing: Each machine's unplanned tickets spread uniformly at random over its own time in the period [exact calculation; Monte Carlo check].

Ticket–telemetry link

Case clock and **telemetry clock:** Case times are US Eastern with daylight saving, as IPA confirmed on Oct 13th, and telemetry times are UTC. We convert ticket times to UTC before measuring any window.

Last 10 codes and **comparison window:** The last 10 codes are the 10 most recent distinct non-success codes in the 30 days before the ticket. The comparison window is the same machine's 30 days ending 35 days before the ticket. It is clean when the machine has no other ticket 5 to 67 days before the ticket.

Paired tickets and **main group:** Paired tickets are the 3,259 device-matched unplanned tickets with a clean comparison window, both windows inside the telemetry period, and neither before the machine's first transaction. The association study's main group drops the 309 with another ticket in the 5 days before or a window outside the machine's telemetry, leaving 2,950.

Door and **electronics codes:** The midterm's provisional groups: door codes 15, 20, 26, 27, 30, 41–48, 56–59, 79, 80, and 81, and electronics codes 05, 13, and 25. Door-named codes 02, 11, 19, 31, 60, 61, and 78 are in other groups.

Code set, own codes, and **crosswalk:** A code set is the clearView codes the crosswalk assigns to a Category, and a ticket's own codes are its Category's set. Three independent drafters (separate AI agents) wrote the crosswalk without seeing any ticket-to-code data, and we froze it on Oct 22nd. It stays provisional until IPA confirms the code meanings.

Other faults' tickets: Tickets whose own set shares no code with the set tested, on types that log it.

3.2 times the odds: Only main-group pairs on types that log the set, with the set in the final 7 days of exactly one window, count. The ratio is pooled over the five sets that could be tested (SAM, User Door, Deposit Door, Main Door, and Computer Board), whose own tickets have at least 10 such pairs and both parts of the ratio non-zero [Mantel–Haenszel; one-sided permutation test, $p = 0.0005$; cluster bootstrap]. Each set's ratio is descriptive; Main Door's interval starts at 1.01, and Computer Board's includes 1.

Sensitivity checks: Planned re-runs with other populations and windows. The pooled 7-day ratio stays between 2.33 and 3.55, with its interval above 1, in every check run so far that gives one.

Scrubex machines: Machines on clearView telemetry types named scrubex, which carry 1,402 of the 1,439 tickets whose Category has a code set their type logs (97%).

IPA's follow-up questions

Request group: One of ten groups fixed in advance: Monitoring for any monitoring ticket, and otherwise a group set from Trouble Code and Category by a lookup table (Electronics, Network, Software, Operator error, Door, Mechanical, Other, No Trouble Found, or Not categorized).

Within model and **share ratio:** Within-model comparisons use the 15 models with at least 100 installed machine-months in each age group. A group's share ratio is its share of a model's young-machine tickets in the nine groups other than Not categorized ÷ its share of the model's older-machine tickets in those groups, pooled over the 15 models [Mantel–Haenszel].

Tests and **descriptive results:** We tested the within-model mix of request groups first [permutation test], then each group [Benjamini–Hochberg], and Mechanical and Electronics pass. Category-level results, such as Touchscreen's, are descriptive, because a preview of Category shares by age was seen while planning.

Extra tickets and **the 80%:** Extra tickets are young machines' tickets minus their machine-months × the older machines' rate. Young machines running above older ones of the same model give 1,156 of the 1,447 (80%), and model mix gives most of the rest.

Install-year cohorts: The install years 2016–18, 2019–21, 2022–23, and 2024–26, on the 15 comparison models. We chose them and found the 2022 step after seeing the data, so both are descriptive.

Repeat classes: Each repeat takes the first class that applies: a monitoring ticket involved, No Trouble Found involved, a blank Category, the same Category, the same request group, or a different group.

Clearly stands out: A model's or telemetry type's observed ÷ random-timing repeats, divided by the fleet's 1.53, has a 95% interval that excludes 1 [cluster bootstrap] and survives correction for testing 12 models or 11 types [Benjamini–Hochberg $q < 0.05$].

References

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[2] Replication notebook 02: High-ticket machines. Jupyter notebook, `https://github.com/zach-yaninek/analytics-lab/blob/c3a95da/notebooks/02_high_ticket_machines.ipynb`, 2026.

[3] Replication notebook 03: Repeat tickets and repeat-contact pairings. Jupyter notebook, `https://github.com/zach-yaninek/analytics-lab/blob/c3a95da/notebooks/03_repeat_tickets.ipynb`, 2026.

[4] Replication notebook 04: Fleet rates, customers vs. their fleets, and persistence. Jupyter notebook, `https://github.com/zach-yaninek/analytics-lab/blob/c3a95da/notebooks/04_fleet_rates_customers_persistence.ipynb`, 2026.

[5] Replication notebook 05: Ticket clock and fields. Jupyter notebook, `https://github.com/zach-yaninek/analytics-lab/blob/c3a95da/notebooks/05_ticket_clock_and_fields.ipynb`, 2026.

[6] Replication notebook 06: First look at codes and the association study. Jupyter notebook, `https://github.com/zach-yaninek/analytics-lab/blob/c3a95da/notebooks/06_telemetry_codes_and_association.ipynb`, 2026.

[7] Replication notebook 07: Request types by age and Electronics cases. Jupyter notebook, `https://github.com/zach-yaninek/analytics-lab/blob/c3a95da/notebooks/07_request_types_and_electronics.ipynb`, 2026.

[8] Replication notebook 08: Verification. Jupyter notebook, `https://github.com/zach-yaninek/analytics-lab/blob/c3a95da/notebooks/08_verification.ipynb`, 2026.